Knowledge Report

Document: java-unit1.pdf

Summary

They may represent a person, place, bank account, table of data, or any other item the program handles. A person can have dif ferent characteristics at the same time (e.g., employee, father , husband, and son). Avoiding Ambiguity: It's commonly used to disambiguate between instance variables and parameters with the same name in constructors or methods. This method performs final operations or cleanup tasks on an object before it is removed from memory . protectedvoidfinalize(){ Cleanupcode } In this example, two objects of the Garbage class are created and then set to null.

Global Key Terms

method | class | system | println | int | java | methods | data | object | example

Page-Wise Insights

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- Global variables allow unrestricted data access across functions, making code dif ficult to maintain and extend.
- Modular Design : Programs are divided into objects based on real-world entities.
- Each object has its own data, leading to better organization and management of code.
- Changes in one part of the program have minimal impact on other parts.

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- Class Example:Allows for overloading of operators, providing flexibility and expressiveness in code.

- Object-oriented programming provides a structured and ef ficient approach to program design.
- They may represent a person, place, bank account, table of data, or any other item the program handles.
- Objects also represent user-defined data such as structures, time, and lists (e.g. Vector).

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- It emphasizes only sharing relevant functionality while hiding lower-level details.
- Interface : A set of methods that classes must implement.
- Encapsulation is the wrapping up of data and functions into a single unit.
- A person can have dif ferent characteristics at the same time (e.g., employee, father , husband, and son).

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- Overloading occurs when two or more methods in the same class have the same name but dif ferent parameters.
- Dynamic binding means the code associated with a procedure call is known only at the time of execution.
- Message passing is a process or type of communication between processes.
- classAnimal{ voidspeak(){ System.out.println("Animal is speaking"); } } classDogextendsAnimal{ @Override voidspeak(){ System.out.println("Dog is barking"); } } In method, the method speak() will exhibit dynamic binding: publicstaticvoidmain(String[] args){ Animal animal =newDog(); animal.speak(); Output: Dog is barking }

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- It allows developers to focus on the essential features of an object while hiding unnecessary details.
- Inheritance reduces redundancy and simplifies code reuse by allowing the extension of existing classes.
- Methods can be overridden in derived classes to provide specific implementations for different types.
- Each module focuses on a specific task or functionality , reducing overall complexity

and improving maintainability .

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- It provides a clear interface while concealing the underlying complexity .
- They of fer reusable solutions to common programming problems, improving code quality and ef ficiency .
- Objects cannot be created directly from an abstract class.
- Abstract methods are typically used when an action is common, but implementations may dif fer.

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- Interface Example: Notes on Classes and Objects In the example provided, the Bike class is an abstract class with an abstract method run().
- Methods in an interface are by default public and abstract.
- Variables in an interface are by default public, static, and final.
- Java was developed by James Gosling and his team at Sun Microsystems in the early 1990s.

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- Java is a programming language created by James Gosling from Sun Microsystems in 1991.
- It is a class-based, object-oriented, simple, and high-level general-purpose language.
- JRE (Java Runtime Environment) A subset of JDK, including JVM and libraries.
- It has a rich set of APIs (Application Protocol Interface).

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- InterpreterJava supports object-oriented concepts such as objects, classes, data abstraction, encapsulation, inheritance, and polymorphism.
- Byte code can be executed on multiple platforms such as Windows, Linux, macOS, and Ubuntu.
- Java is more secure compared to other languages such as C and C++.
- Java supports platform-independent and architecture-neutral features, making it portable across dif ferent systems

portable across different systems.

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- The JIT compiler is enabled by default.
- Java supports multi-threading, allowing multiple programs to execute concurrently and simultaneously .
- Java enables users to create distributed applications.
- Technologies such as RMI (Remote Method Invocation) and Enterprise JavaBeans are used for creating distributed applications.

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- Flowchart for Data Types and Its Subtypes Description of Data Types Data TypeBytes Description byte 1 Integer values from -128 to 127 short 2 Integer values from -32768 to 32767 int 4 Integer values from -2147483648 to 2147483647 long 8 Integer values from -9223372036854775808 to 9223372036854775807 float 4 Single-precision floating-point numbers double 8 Double-precision floating-point numbers boolean 1 Boolean values (true or false) char 2 Unicode character (single character) 2.
- Non-Primitive Data Typesboolean Created by programmers, also known as reference variables or object references.
- Variables reference the memory location where data is stored.
- Examples include arrays, classes, and interfaces.

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- Local V ariables Declared inside the body of a method.
- Should be declared and initialized before the first use.
- Class V ariables (Static V ariables) Declared with the static keyword in a class but outside a method.
- Example: double pi =3.14; Compile-time initialization double area = pi *1.5; Runtime initialization The scope of a variable refers to the area or section of a program where the variable can be accessed.

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- Lifetime of a Variable Arrays and Operators

ArraysclassDemoScope{ publicargs){int x =10; System.out.Scope is within the main method if(x 10) { int y =20; Scope is within the 'if' block System.out.println("x and y: "+ x +" "+ y); } x = x +2; System.out.println(y); Compile-time error: Cannot access 'y' outside of 'if' block System.out.println("x is "+ x); } } In the example, the variable y is declared inside the if block, so its scope is limited to that block.

- It cannot be accessed outside the block.
- An array is a linear data structure.
- Arrays are used to store multiple values in a single variable.

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- Arrays in Java can store both primitive and non-primitive types of data.
- Individual values in an array are called elements, and the elements are stored in contiguous memory locations.
- 1-Dimensional Array Uses a single index to store elements.
- Syntax: datatype[][] arrayName = new datatype[x][y]; Example: int a[][]={{21,22,23}, {54,54,14}}; for(int i =0; i <2; i){ for(int j =0; j <3; j){ System.out.print(a[i][j]+" "); } System.out.println(); } Output: 21 22 23 54 54 14 You can perform addition of two matrices using arrays.

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- Expressions: Expressions are combinations of variables, operators, and method calls that evaluate to a single value.
- They can be used to perform calculations or make decisions in Java programs.
- Types of Expressions{4,4,5} }; intB[][]={ {11,2,3}, {13,15,4}, {23,25,4} }; intC[][]=newint[3][3]; for(int i =0; i <3; i){ for(int j =0; j <3; j){ C[i][j]=A[i][j]+B[i][j]; System.out.print(C[i][j]+" "); } System.out.println(); } } } In this example, two 3x3 matrices are defined (A and B), and their sum is stored in matrix C. The resulting matrix C is printed.
- Operator : A symbol that performs tasks such as arithmetic, logical, and conditional operations.